

6-Month SQL + Power BI + Excel Learning Roadmap & Checklist

Goal: Become job-ready Data Analyst in 6 months

Outcome: Strong SQL + Power BI projects + interview confidence

◆ Month 1 – SQL Basics & Power BI Foundation

Week 1 – SQL Fundamentals

- ☐ Learn **SELECT, WHERE, ORDER BY, LIMIT**
- ☐ Understand tables, rows, columns, and data types
- ☐ Practice basic queries on SQLZoo / Mode

Week 2 – SQL Aggregations

- ☐ Learn **GROUP BY**
- ☐ Practice **SUM, AVG, COUNT, MIN, MAX**
- ☐ Solve **10 beginner SQL problems** (HackerRank / LeetCode)

Week 3 – Power BI Setup

- ☐ Install Power BI Desktop
- ☐ Import CSV / Excel data
- ☐ Explore visuals (bar, line, pie, table)
- ☐ Understand fields pane & visual pane

Week 4 – Mini Project

- ☐ Write SQL to calculate **sales by category**
 - ☐ Load SQL output into Power BI
 - ☐ Create first dashboard with filters
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◆ Month 2 – SQL Joins & Power Query

Week 1 – SQL Joins

- ☐ Learn **INNER**, **LEFT**, **RIGHT**, **FULL JOIN**
- ☐ Combine Customers + Orders tables
- ☐ Understand join use-cases for business data

Week 2 – Joins + Aggregation

- ☐ Practice **JOIN** + **GROUP BY**
- ☐ Calculate KPIs:
 - ☐ Total Sales
 - ☐ Average Order Value
 - ☐ Orders per Customer

Week 3 – Power Query

- ☐ Clean data using Power Query
- ☐ Remove duplicates
- ☐ Handle null values
- ☐ Merge & append tables

Week 4 – Project

- ☐ Build **multi-table Power BI dashboard**
 - ☐ Add slicers (date, region, category)
 - ☐ Add KPI cards
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◆ Month 3 – Subqueries, CTEs & Data Modeling

Week 1 – Subqueries & CTEs

- ☐ Write subqueries using **IN**, **EXISTS**
- ☐ Create CTEs using **WITH**
- ☐ Break complex SQL into readable logic

Week 2 – SQL Functions

- ☐ Use **CASE WHEN**
- ☐ Practice string functions
- ☐ Practice date functions
- ☐ Solve **10 intermediate SQL problems**

Week 3 – Data Modeling

- ☐ Understand **Fact vs Dimension tables**
- ☐ Design a **Star Schema**
- ☐ Create relationships in Power BI Model view

Week 4 – Project

- ☐ Build **Customer Segmentation Dashboard**
 - ☐ Use RFM logic (Recency, Frequency, Monetary)
 - ☐ Document data model & assumptions
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◆ Month 4 – Window Functions & DAX

Week 1 – SQL Window Functions

- ☐ Learn **ROW_NUMBER, RANK**
- ☐ Use **SUM() OVER()**
- ☐ Calculate running totals & rankings

Week 2 – DAX Fundamentals

- ☐ Difference between **Calculated Column vs Measure**
- ☐ Write basic measures:
 - ☐ **SUM()**
 - ☐ **COUNT()**
 - ☐ **AVERAGE()**

Week 3 – Time Intelligence

- ☐ Create YTD measures
- ☐ Create YoY comparison
- ☐ Monthly growth calculations

Week 4 – Project

- ☐ Build **Time-Series Sales Dashboard**
 - ☐ Combine SQL + DAX logic
 - ☐ Add trend & variance analysis
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◆ **Month 5 – Advanced SQL & Power BI Service**

Week 1 – Advanced SQL

- ☐ Learn **PIVOT**
- ☐ Recursive CTEs (basics)
- ☐ Query optimization fundamentals

Week 2 – Advanced DAX

- ☐ Use **CALCULATE()**
- ☐ Use **FILTER()**
- ☐ Use **SUMMARIZE()**
- ☐ Understand context transition

Week 3 – Power BI Service

- ☐ Publish report to Power BI Service
- ☐ Configure scheduled refresh
- ☐ Understand workspace & sharing

Week 4 – Capstone Project

- ☐ Build **End-to-End E-commerce Analytics Dashboard**
 - ☐ Business KPIs + insights
 - ☐ Presentation-ready visuals
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◆ **Month 6 – Portfolio & Job Preparation**

Week 1 – Portfolio Setup

- ☐ Finalize **3 Power BI dashboards**
- ☐ Organize SQL scripts
- ☐ Upload projects to GitHub

Week 2 – Interview Preparation

- ☐ Practice SQL interview questions (LeetCode)
- ☐ Revise joins, CTEs, window functions
- ☐ Prepare DAX interview scenarios

Week 3 – Certification

- ☐ Study for **PL-300 Power BI Data Analyst**
- ☐ Revise Power Query, DAX, Modeling

Week 4 – Job Applications

- ☐ Update resume with project impact
 - ☐ Optimize LinkedIn headline & summary
 - ☐ Apply to **10 Data Analyst roles per week**
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Recommended Learning Resources

Power BI

1 Microsoft Learn (Official)

<https://learn.microsoft.com/en-us/training/powerplatform/power-bi/>

- ☒ Free | Best structured learning path

2 SQLBI

<https://www.sqlbi.com/>

- ☒ Advanced DAX & modeling concepts

3 Enterprise DNA

<https://enterprisedna.co/>

- ☒ Real-world dashboards & case studies
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SQL

4 Mode SQL Tutorial

<https://mode.com/sql-tutorial/>

- ☒ Hands-on analytics-focused SQL

5 LeetCode SQL

<https://leetcode.com/problemset/database/>

✓ Best for interview-level SQL

Interview Preparation (YouTube)

 Rajat Jain – Data Analytics & SQL Interviews

<https://www.youtube.com/@rajatjain001>

Check Other Resources and Next Live Workshops

https://topmate.io/rajat_jain01/